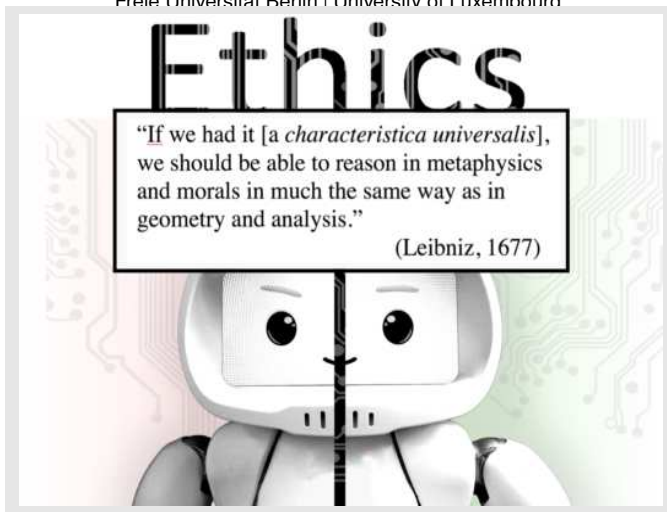


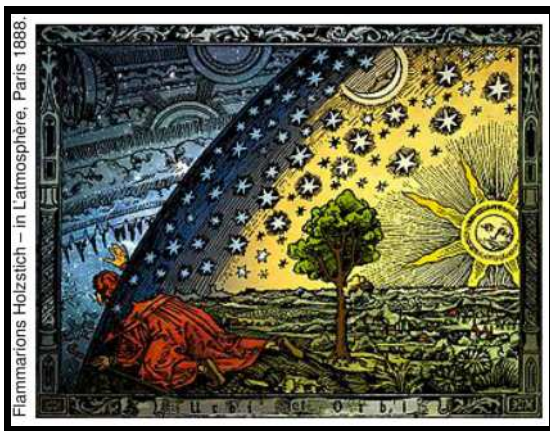
Studies in Computational Metaphysics & Computational (Pseudo-)Ethics

Christoph Benzmüller

Freie Universität Berlin | University of Luxembourg



UNICAMP — July 26, 2019



— Computational Metaphysics — Ontological Argument on the Computer

Related work (on earlier variants):

- ▶ Ed Zalta (& co) with PROVER9 at Stanford [AJP 2011, CADE 2015]
- ▶ John Rushby with PVS at SRI [CAV-WS 2013, JAL 2018]

Ontological Argument — A Long Tradition



Anselm v. Canterbury (1033-1109)

Rational Argument for Existence of God

God is ... that being than which nothing greater can be conceived.

(Proslogion, 1077/78)

It follows:

God exists!?

Anselm v. C.
Gaunilo

Th. Aquinas

Descartes
Spinoza
Leibniz

Hume
Kant

Hegel

Frege

Hartshorne
Malcolm
Lewis
Plantinga
Gödel

Ontological Argument — A Long Tradition



Kurt Gödel (1914-1976) with Einstein

Rational Argument for Existence of God

Definition:

A Godlike being possesses all positive properties.

(plus further axioms and definitions)

Theorem:

God exists necessarily.

Anselm v. C.
Gaunilo

Th. Aquinas

Descartes
Spinoza
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Frege

Hartshorne
Malcolm
Lewis
Plantinga
Gödel

“God is dead.”

- Nietzsche, 1883

“Nietzsche is dead.”

- God, 1900

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Computational Metaphysics: Gödel's Ontological Argument (1970)

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Computational Metaphysics: Gödel's Ontological Argument (1970)



Notion of “Godlike”:

- ▶ Being Godlike is equivalent to having all positive properties.

Note: this definition is “second-order”.

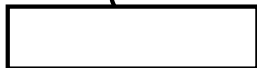
Computational Metaphysics: Gödel's Ontological Argument (1970)

In the end we prove

- ▶ **Necessarily (N)**, there exists God.

Note: we need to formalize “necessity” and “possibility”.

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Computational Metaphysics: Gödel's (1970) & Scott's Variants (1972)

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Computational Metaphysics: Gödel's (1970) and Scott's Variants (1972)



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(Main) Difference between Gödel and Scott: Def. of “Essence (Ess.)”

Gödel: Property E is Ess. of x iff all of x's properties are nec. entailed by E.

Scott: Property E is Ess. of x iff **x has E and** all of x's properties are nec. entailed by E.

Computational Metaphysics: Dana Scott's Variant

Axiom Either a property or its negation is positive, but not both: $\forall\phi[P(\neg\phi) \leftrightarrow \neg P(\phi)]$

Axiom A property necessarily implied by a positive property is positive:
 $\forall\phi\forall\psi[(P(\phi) \wedge \Box\forall x[\phi(x) \rightarrow \psi(x)]) \rightarrow P(\psi)]$

Thm. Positive properties are possibly exemplified: $\forall\phi[P(\phi) \rightarrow \Diamond\exists x\phi(x)]$

Def. A **Godlike** being possesses all positive properties: $G(x) \leftrightarrow \forall\phi[P(\phi) \rightarrow \phi(x)]$

Axiom The property of being Godlike is positive: $P(G)$

Cor. Possibly, God exists: $\Diamond\exists xG(x)$

Axiom Positive properties are necessarily positive: $\forall\phi[P(\phi) \rightarrow \Box P(\phi)]$

Def. An essence of an individual is a **property possessed by it and** necessarily implying any of its properties:
 $\phi \text{ ess. } x \leftrightarrow \phi(x) \wedge \forall\psi(\psi(x) \rightarrow \Box\forall y(\phi(y) \rightarrow \psi(y)))$

Thm. Being Godlike is an essence of any Godlike being: $\forall x[G(x) \rightarrow G \text{ ess. } x]$

Def. *Necessary existence* of an individual is the necessary exemplification of all its essences:
 $NE(x) \leftrightarrow \forall\phi[\phi \text{ ess. } x \rightarrow \Box\exists y\phi(y)]$

Axiom Necessary existence is a positive property: $P(NE)$

Thm. Necessarily, God exists: $\Box\exists xG(x)$



Computational Metaphysics: Scott's Variant

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Thm. Necessarily, God exists: $\Box \exists x G(x)$

Difference to Gödel (who omits this conjunct)

Variant of Kurt Gödel

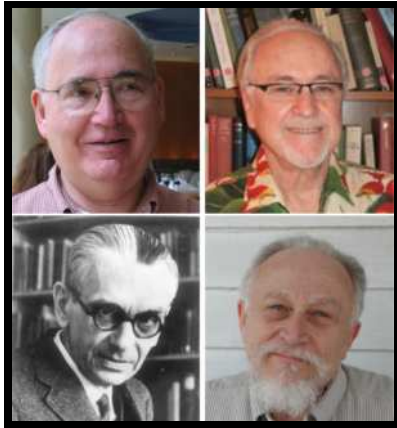
- ▶ premises are **inconsistent/contradictory**
- ▶ Philosophers had not seen this ... but my theorem prover LEO-II did

Variant of Dana Scott

- ▶ premises are **consistent**
- ▶ all steps **logically correct** already in higher-order modal logic **KB**

Variant of Dana Scott: further corollaries

- ▶ Monotheism
- ▶ Gott is flawless (has only positive properties)
- ▶ **Modal collapse:** $\varphi \rightarrow \Box \varphi$
 - ▶ there are no contingent truths
 - ▶ no alternative worlds
 - ▶ everything is determined
 - ▶ no free will



$$\varphi \rightarrow \Box \varphi$$

How to avoid the Modal Collapse?

- ▶ **Godlike** has been defined in terms of **positive properties**
- ▶ **positive properties** has been linked with the notion of **ultrafilter**
- ▶ In our experiments we then distinguished between
 - $\mathcal{P}$: positive **intensional properties**
 - $\mathcal{P}'$: positive **extensions of properties** (rigidly intensionalised extensions)
- ▶ Gödel/Scott variant axiomatises $\mathcal{P}$: $\mathcal{P} = \mathcal{P}'$ is an **ultrafilter**
- ▶ Anderson's variant axiomatises $\mathcal{P}$: $\mathcal{P} \neq \mathcal{P}'$; only $\mathcal{P}'$ is an **ultrafilter**
- ▶ Fitting's variant axiomatises only $\mathcal{P}'$: $\mathcal{P}'$ is an **ultrafilter**

Modal collapse holds for Gödel/Scott variant, but not for Anderson's & Fitting's!

They achieve this in seemingly different ways.

Mathematically, however, their solutions appear closely related.

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“If we had it [a *characteristica universalis*], we should be able to reason in metaphysics and morals in much the same way as in geometry and analysis.”

(Leibniz, 1677)

**Flexible Reasoning Technology:
Shallow Semantical Embeddings (SSEs)
in Classical Higher-Order Logic (HOL)**



STUDIES IN LOGIC
AND
PRACTICAL REASONING

VOLUME 3

D.M. GABBAY / P. GARDENFORS / J. SIEKMANN / J. VAN BENTHEM / M. VARDI / J. WOODS

EDITORS

*Handbook of
Modal Logic*

2 BASIC MODAL LOGIC

In this section we introduce the basic modal language and its relational semantics. We define basic modal syntax, introduce models and frames, and give the satisfaction definition. We then draw the reader's attention to the internal perspective that modal languages offer on relational structure, and explain why models and frames should be thought of as graphs. Following this we give the standard translation. This enables us to convert any basic modal formula into a first-order formula with one free variable. The standard translation is a bridge between the modal and classical worlds, a bridge that underlies much of the work of this chapter.

2.1 First steps in relational semantics

Suppose we have a set of proposition symbols (whose elements we typically write as p, q, r and so on) and a set of modality symbols (whose elements we typically write as m, m', m'' , and so on). The choice of PROP and MOD is called the *signature* (or *similarity type*) of the language; in what follows we'll tacitly assume that PROP is denumerably infinite, and we'll often work with signatures in which MOD contains only a single element. Given a signature, we define the *basic modal language* (over the signature) as follows:

$$\varphi ::= p \mid \top \mid \perp \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \rightarrow \psi \mid \varphi \leftrightarrow \psi \mid \langle m \rangle \varphi \mid [m] \varphi.$$

That is, a basic modal formula is either a proposition symbol, a boolean constant, a boolean combination of basic modal formulas, or (most interesting of all) a formula prefixed by a diamond

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2.1 First steps in relational semantics

Syntax

Metalanguage

What follows we tacitly assume that $PROP$ is countably infinite, and we'll often work with signatures in which MOD contains only a single element. Given a signature, we define the *basic modal language* (over the signature) as follows:

$$\varphi ::= p \mid \top \mid \perp \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \rightarrow \psi \mid \varphi \leftrightarrow \psi \mid \langle m \rangle \varphi \mid [m] \varphi.$$

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Example: Modal Logic Textbook

A *model* (or *Kripke model*) $\mathfrak{M}$ for the basic modal language (over some fixed signature) is a triple $\mathfrak{M} = (W, \{R^m\}_{m \in \text{MOD}}, V)$. Here W , the *domain*, is a non-empty set, whose elements we usually call *points*, but which, for reasons which will soon be clear, are sometimes called *states*, *times*, *situations*, *worlds* and other things besides. Each R^m in a model is a binary relation on W , and V is a function (the valuation) that assigns to each proposition symbol p in PROP a subset $V(p)$ of W ; think of $V(p)$ as the set of points in $\mathfrak{M}$ where p is true. The first two components $(W, \{R^m\}_{m \in \text{MOD}})$ of $\mathfrak{M}$ are called the *frame* underlying the model. If there is only one relation in the model, we typically write (W, R) for its frame, and (W, R, V) for the model itself. We encourage the reader to think of Kripke models as graphs (or to be slightly more precise, *directed graphs*, that is, graphs whose points are linked by directed arrows) and will shortly give some examples which show why this is helpful.

Suppose w is a point in a model $\mathfrak{M} = (W, \{R^m\}_{m \in \text{MOD}}, V)$. Then we inductively define the notion of a formula φ being *satisfied* (or *true*) in $\mathfrak{M}$ at point w as follows (we omit some of the clauses for the booleans):

$\mathfrak{M}, w \models p$	iff	$w \in V(p)$,
$\mathfrak{M}, w \models \top$		always,
$\mathfrak{M}, w \models \perp$		never,
$\mathfrak{M}, w \models \neg\varphi$	iff	not $\mathfrak{M}, w \models \varphi$ (notation: $\mathfrak{M}, w \not\models \varphi$),
$\mathfrak{M}, w \models \varphi \wedge \psi$	iff	$\mathfrak{M}, w \models \varphi$ and $\mathfrak{M}, w \models \psi$,
$\mathfrak{M}, w \models \varphi \rightarrow \psi$	iff	$\mathfrak{M}, w \not\models \varphi$ or $\mathfrak{M}, w \models \psi$,
$\mathfrak{M}, w \models \langle m \rangle \varphi$	iff	for some $v \in W$ such that $R^m wv$ we have $\mathfrak{M}, v \models \varphi$,
$\mathfrak{M}, w \models [m] \varphi$	iff	for all $v \in W$ such that $R^m wv$ we have $\mathfrak{M}, v \models \varphi$.

Example: Modal Logic Textbook

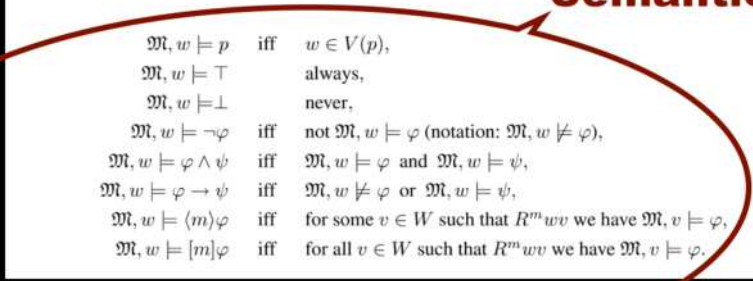
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Metalanguage

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Semantics



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$\mathfrak{M}, w \models [m] \varphi$	iff	for all $v \in W$ such that $R^m wv$ we have $\mathfrak{M}, v \models \varphi$.

Kripke Style Semantics

$M, g, s \models P$ if and only if $s \in g(P)$

$M, g, s \models \neg \varphi$ if and only if $M, g, s \not\models \varphi$

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Standard Translation for Propositional Fragment (encoded in HOL)

- ▶ "lifted" predicate $P_{i \rightarrow o}$
- ▶ $\neg_{(i \rightarrow o) \rightarrow (i \rightarrow o)} \varphi_{i \rightarrow o} w_i = \neg(\varphi w)$
- ▶ $\vee_{(i \rightarrow o) \rightarrow (i \rightarrow o) \rightarrow (i \rightarrow o)} \varphi_{i \rightarrow o} \psi_{i \rightarrow o} w_i = \varphi w \vee \psi w$
- ▶ $\Box_{(i \rightarrow o) \rightarrow (i \rightarrow o)} \varphi_{i \rightarrow o} w_i = \forall v_i R w v \rightarrow \varphi v$

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Validity

- ▶ $[\varphi_{i \rightarrow o}] = \forall w_i \varphi w$ resp. $[\cdot] = \lambda \varphi_{i \rightarrow o} \forall w_i \varphi w$ (global)
- ▶ $[\varphi_{i \rightarrow o}]_{cw} = \varphi cw$ resp. $[\cdot]_{cw} = \lambda \varphi_{i \rightarrow o} \varphi cw$ (local)

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Modal Logic (in fact, Hybrid Logic) as a Fragment of HOL

Kripke Style Semantics

$M, g, s \models \forall x \varphi$ if and only if for all $d \in D$ we have $M, ([d/x]g), s \models \varphi$

Standard Translation extended for Quantifiers (and encoded in HOL)

- ▶ in HOL $\forall x_\alpha \varphi x$ is shorthand for $\Pi_{(\alpha \rightarrow o) \rightarrow o} (\lambda x_\alpha \varphi x)$ —no binder needed!!!
- ▶ $\Pi_{(\alpha \rightarrow (i \rightarrow o)) \rightarrow (i \rightarrow o)} = \lambda \Phi_{\alpha \rightarrow (i \rightarrow o)} \lambda w_i \Pi_{(\alpha \rightarrow o) \rightarrow o} (\lambda x_\alpha \Phi x w)$

Example (compositionality and λ -conversion at work)

$$\begin{aligned} [\Box \forall x Px] &\equiv [\Box \Pi(\lambda x Px)] &\equiv [\Box \Pi(\lambda x \lambda w Pxw)] \\ &\equiv [\Box ((\lambda \Phi \lambda w \Pi(\lambda x \Phi xw))(\lambda x \lambda w Pxw))] \\ &\equiv [\Box (\lambda w \Pi(\lambda x (\lambda w Pxw)xw))] \\ &\equiv [\Box (\lambda w \Pi(\lambda x Pxw))] \\ &\equiv [(\lambda \varphi \lambda w \Pi(\lambda v R w v \rightarrow \varphi v))(\lambda w \Pi(\lambda x Pxw))] \\ &\equiv [(\lambda w \Pi(\lambda v R w v \rightarrow (\lambda w \Pi(\lambda x Pxw))v))] \\ &\equiv [\lambda w \Pi(\lambda v R w v \rightarrow \Pi(\lambda x Pxv))] \\ &\equiv [\lambda w \forall v (R w v \rightarrow \forall x Pxv)] \\ &\equiv \forall w \forall v (R w v \rightarrow \forall x Pxv) \end{aligned}$$

- ▶ above: possibilist quantification
- ▶ actualist quantification: $\Pi = \lambda \Phi \lambda w \Pi(\lambda x \text{ existsAt } x \ w \rightarrow \Phi x w)$
- ▶ also supported: local and global validity and consequence

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- ▶ above: possibilist quantification
- ▶ actualist quantification: $\Pi = \lambda \Phi \lambda w \Pi(\lambda x \text{ existsAt } x \ w \rightarrow \Phi x w)$
- ▶ also supported: local and global validity and consequence

Kripke Style Semantics

$M, g, s \models \forall x \varphi$ if and only if for all $d \in D$ we have $M, ([d/x]g), s \models \varphi$

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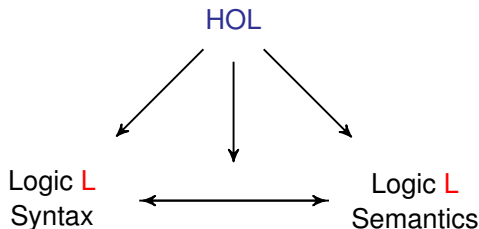
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Flexible Reasoning Technology: SSEs in HOL



Examples for **L** we have already studied:

Intuitionistic Logics, (Mathematical) Fuzzy Logics, Free Logic, Modal Logics, Description Logics, Conditional Logics, Access Control Logics, Hybrid Logics, Multivalued Logics, Logics with Neighborhood Semantics, Paraconsistent Logics, Deontic Logics, ...

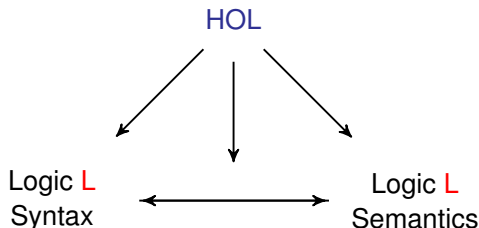
Embedding works also for quantifiers (first-order & higher-order)

HOL provers become universal logic reasoning engines!

interactive: Isabelle/HOL, PVS, HOL4, Hol Light, Coq/HOL, ...

automated: Leo-III, LEO-II, Satallax, TPS, Nitpick, Isabelle/HOL, ...

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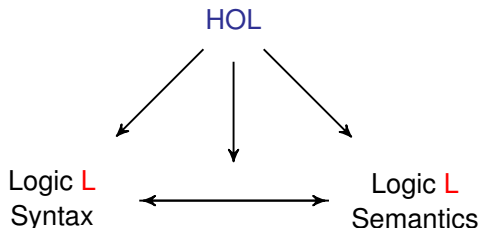
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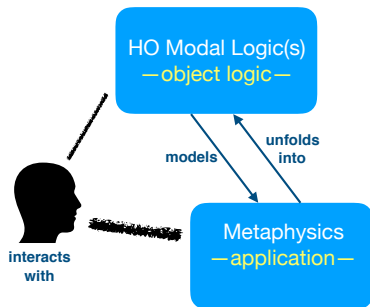
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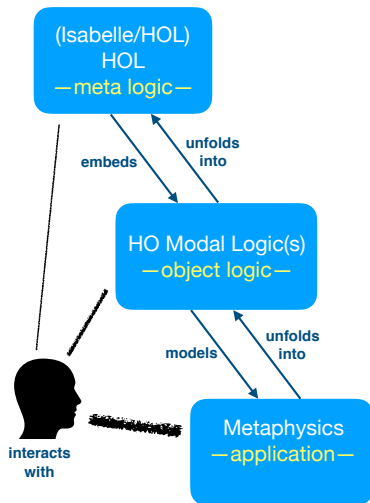
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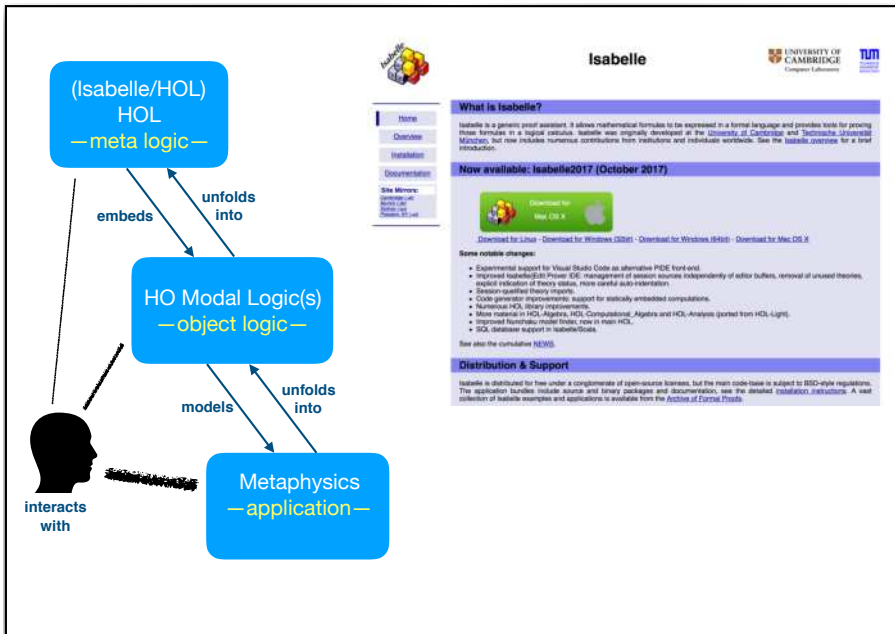
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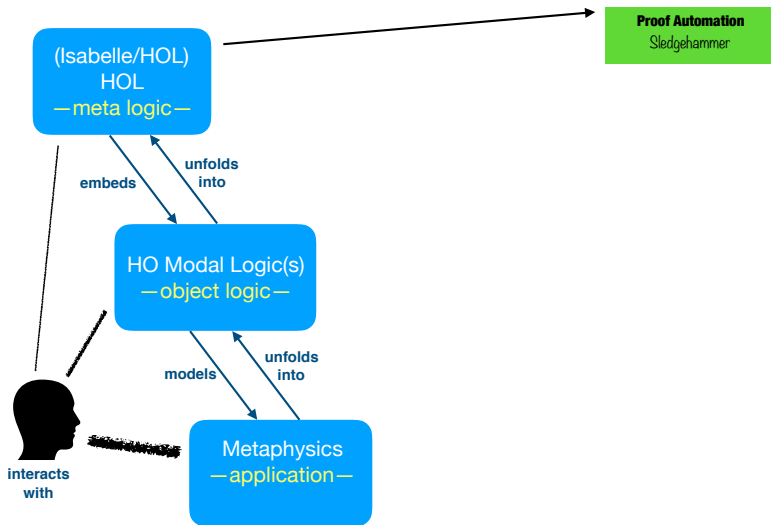


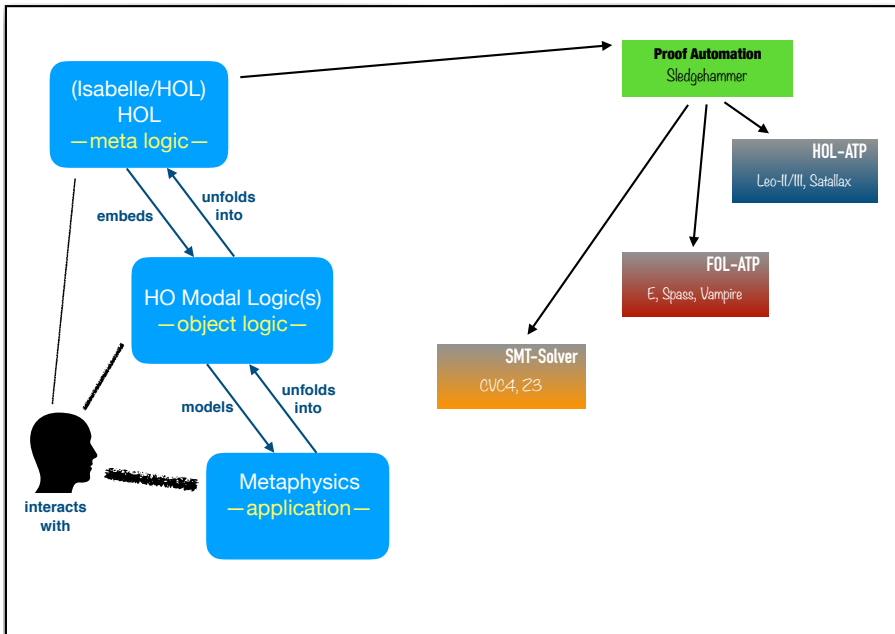


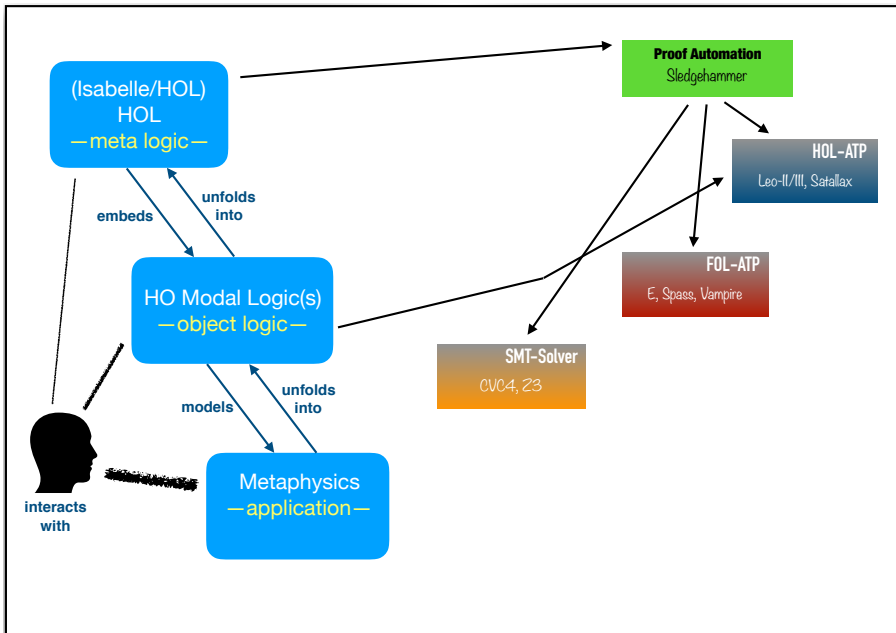


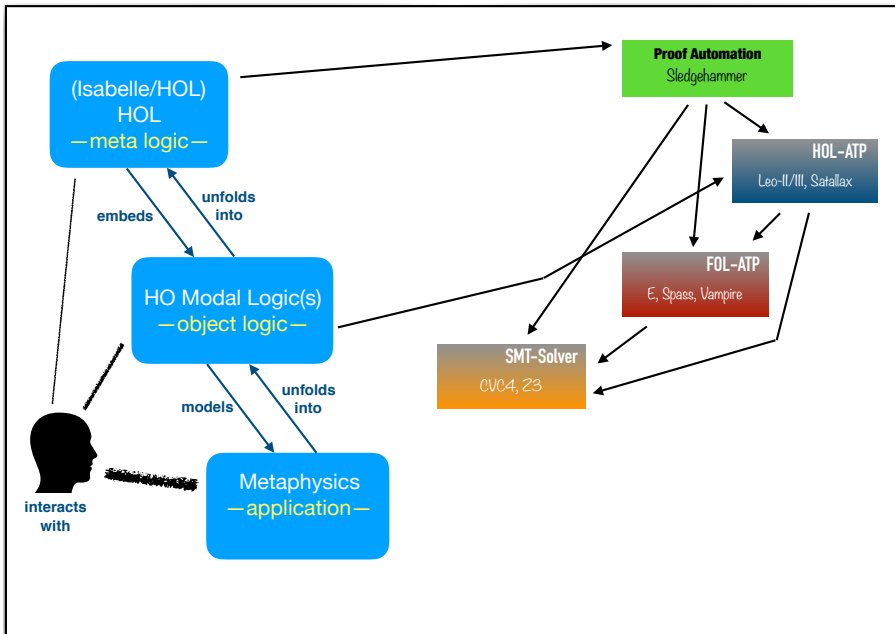


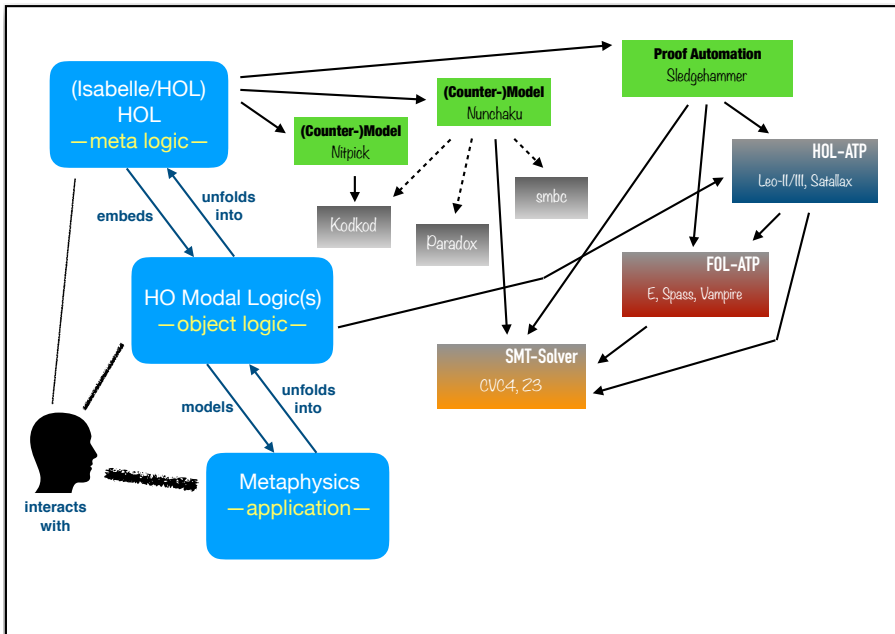
Flexible Reasoning Technology: SSEs in HOL [SCP, 2019], [OpenPhilos., 2019]

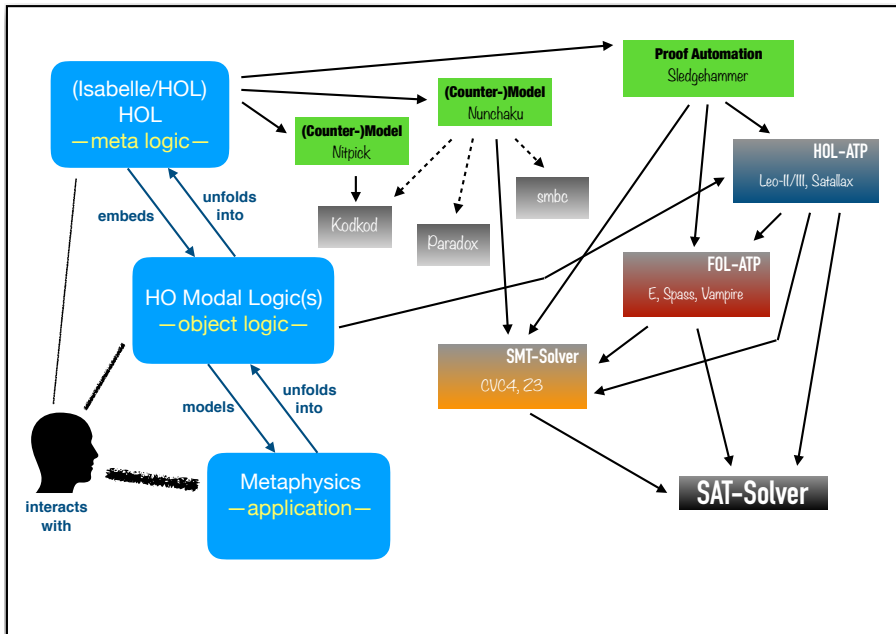


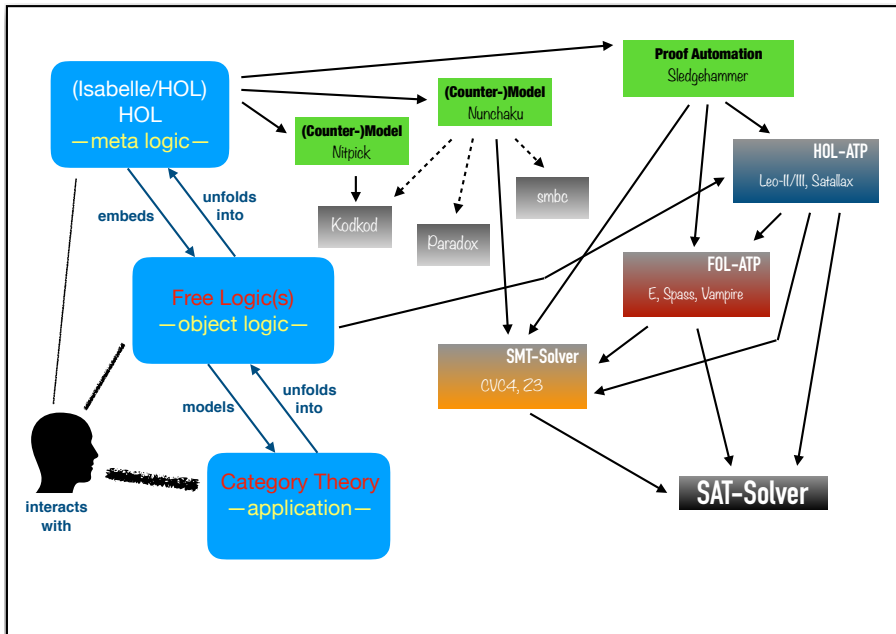


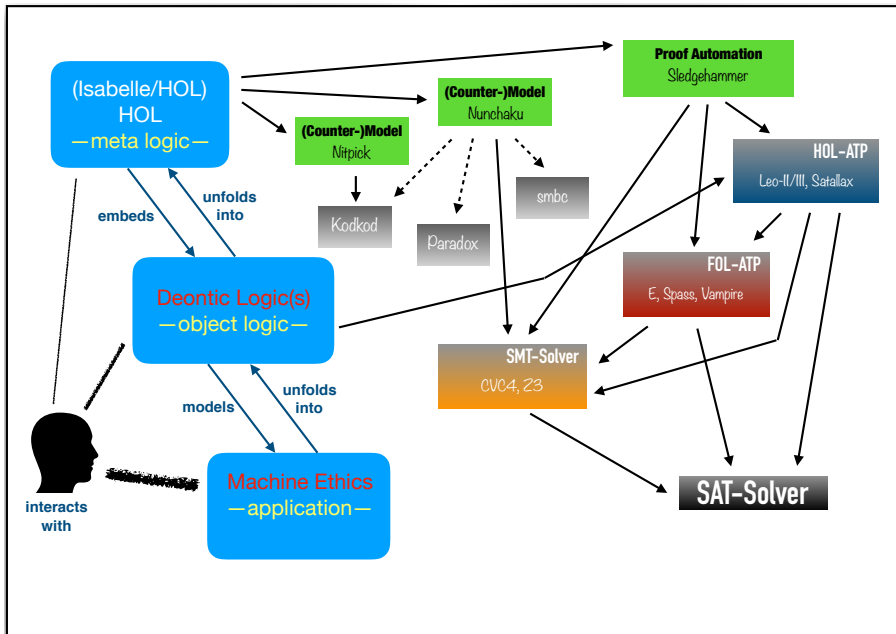


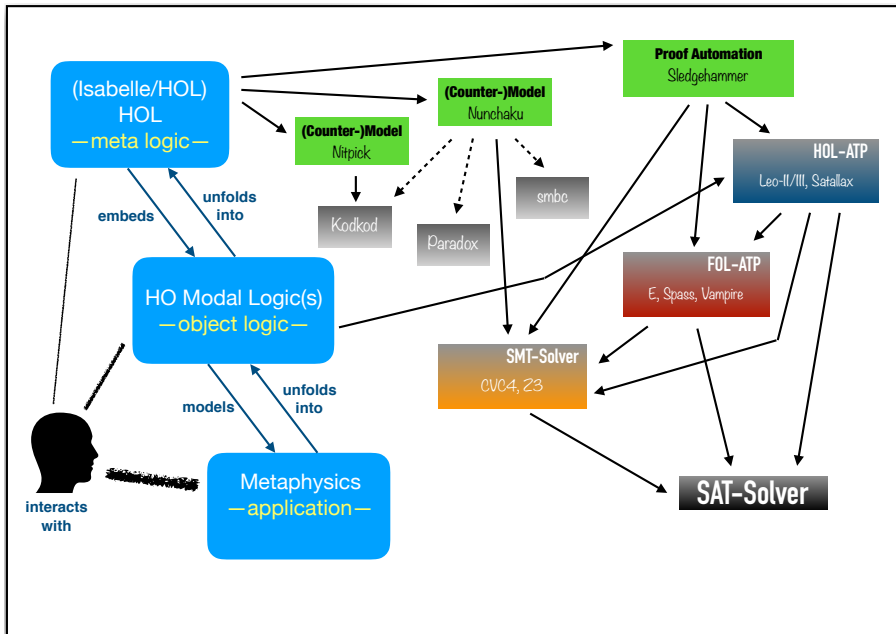












Flexible Reasoning Technology: SSEs in HOL



Ed Zalta (Stanford)

Principia Logico-Metaphysica

Hyperintensional higher-order modal logic

Inconsistency/Paradox detected & fixed

[Open Philosophy, in print] [RSL, in print]



Daniel Kirchner
(Mathematics, FU Berlin)

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Daniel Kirchner
(Mathematics, FU Berlin)

Kirchner Paradox

Daniel & Isabelle/HOL have become close advisors of
Ed Zalta in the search for a repair

Computational Metaphysics par excellence!!!

Flexible Reasoning Technology: SSEs in HOL



Ed Zalta (Stanford)

Principia Logico-Metaphysica

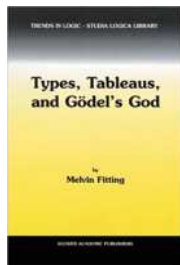
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Daniel Kirchner
(Mathematics, FU Berlin)



M. Fitting (New York)

Philosophy/Metaphysics

Ontological Argument
(avoids modal collapse)

Intensional higher-order modal logic

Verified (main chapters)

[Archive of Formal Proofs, 2017]



David Fuenmayor
(Philosophy, FU Berlin)

Flexible Reasoning Technology: SSEs in HOL



E. J. Lowe's (Durham)

Philosophy/Metaphysics

E. J. Lowe's Modal Ontological Argument

Higher-order modal logic

Verified (minor corrections)

[Journal of Applied Logics, 2018]



David Fuenmayor
(Philosophy, FU Berlin)

Flexible Reasoning Technology: SSEs in HOL



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Verified (minor corrections)

[Journal of Applied Logics, 2018]



David Fuenmayor
(Philosophy, FU Berlin)



P. Freyd & A. Scedrov

Category Theory

Free first-order logic

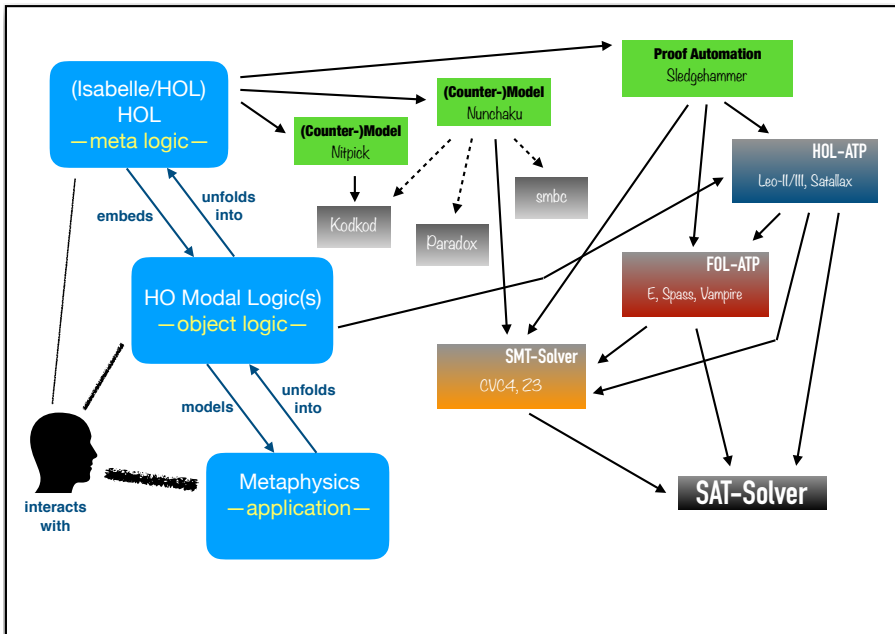
(Constricted) Inconsistency detected & fixed

[JAR, 2019]

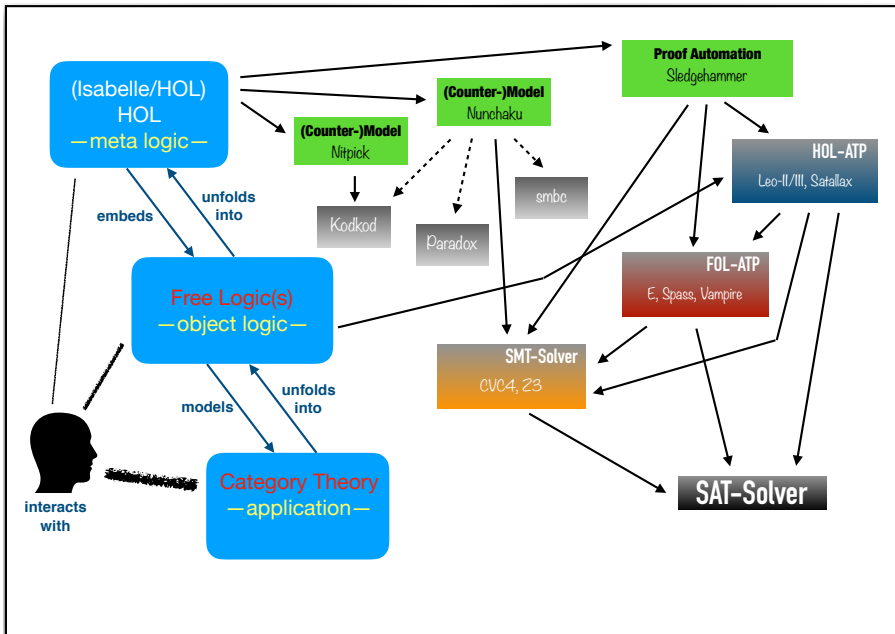


D. Scott
(UC Berkeley)

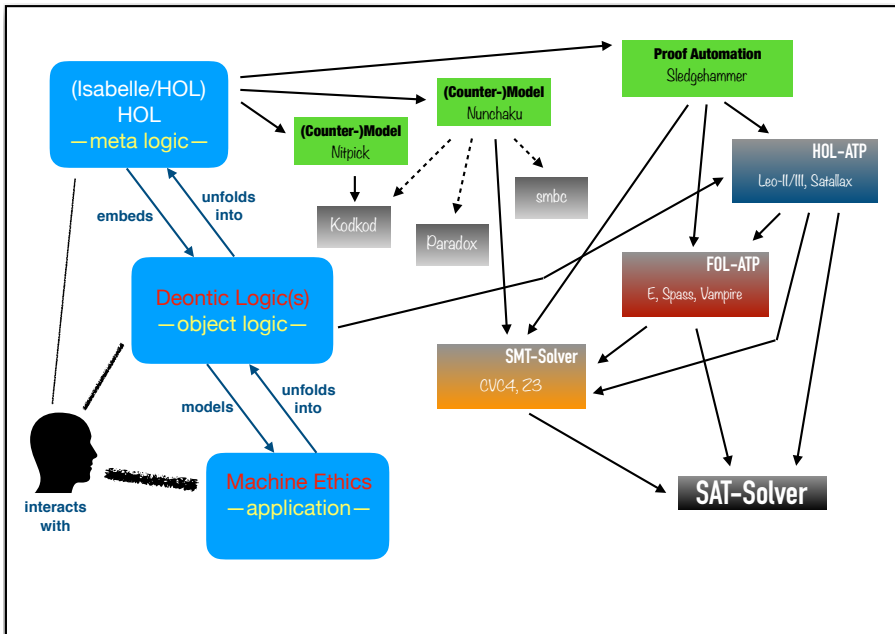
Flexible Reasoning Technology: SSEs in HOL



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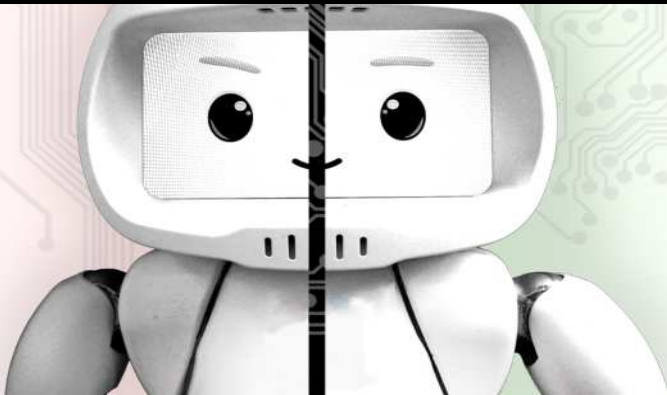
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Ethics

Peaceful coexistence with **intelligent autonomous systems (IASs)**?

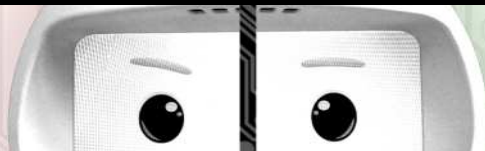
- ▶ appropriate forms of **machine-control**
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Ethics

Peaceful coexistence with **intelligent autonomous systems (IASs)**?

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- ▶ appropriate forms of **human-machine-interaction**



Existing societal processes are based on:

- ▶ **rational argumentation & dialog**
- ▶ **explicit normative reasoning** (legal & ethical)

Deployment of IASs lacking such competencies? How wise is this?

Motivation: (Pseudo-)Ethical Intelligent Systems

Long-term: Emerging Superintelligence

Really? Anyhow ...

- ▶ How to prevent Superintelligence from turning against humanity?

Medium-term: Development of pseudo-ethical skills in IASs

- ▶ Which norms? Which reasoning principles?
- ▶ What architectural design? What functionalities?
- ▶ How to implement, deploy and verify?

Different kinds of systems and approaches:

- ▶ [Moor, 2009]:
 - ethical impact agents (ethical consequences to actions)
 - implicit ethical agents (ethical reactions to given situations)
 - explicit ethical agents (reasoning with ethical theories/rules)
 - full ethical agents (conscious, intentional, free will)
- ▶ bottom-up vs. top-down
- ▶ [DoranEtAl., 2017]:
opaque — comprehensible — interpretable — explainable AI

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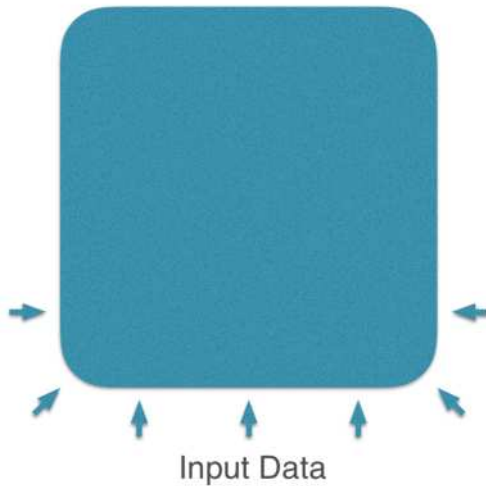
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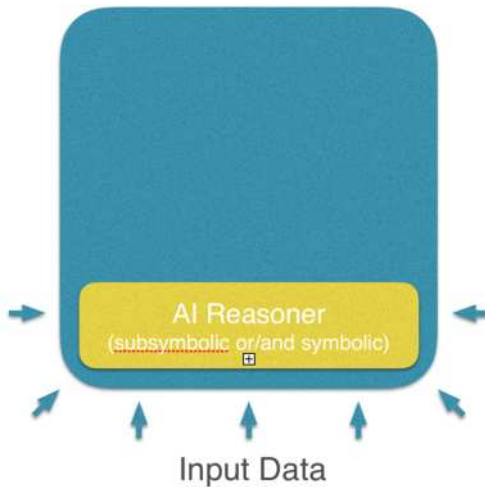
IAS

A teal-colored rounded square with the letters "IAS" in white, centered within it.

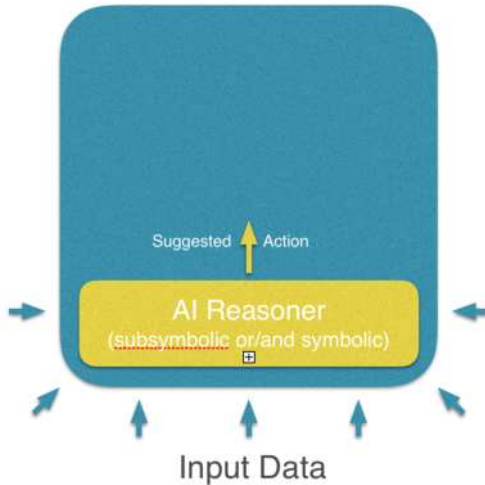
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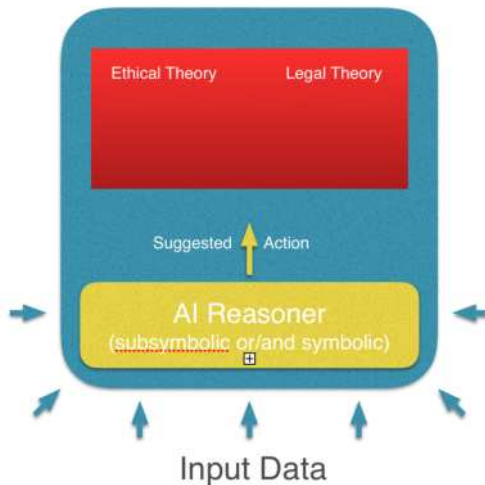
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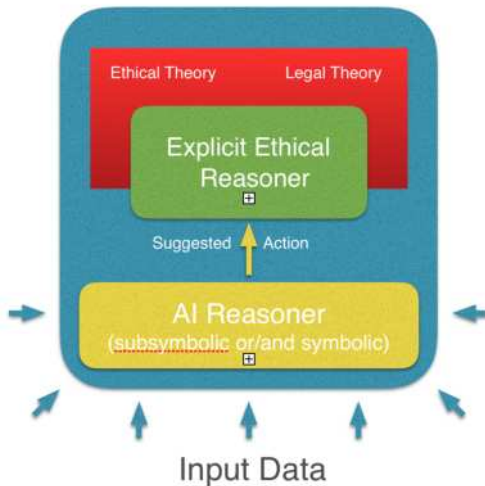
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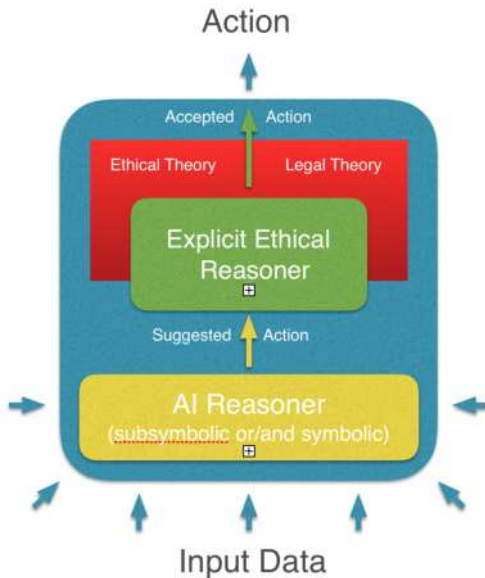
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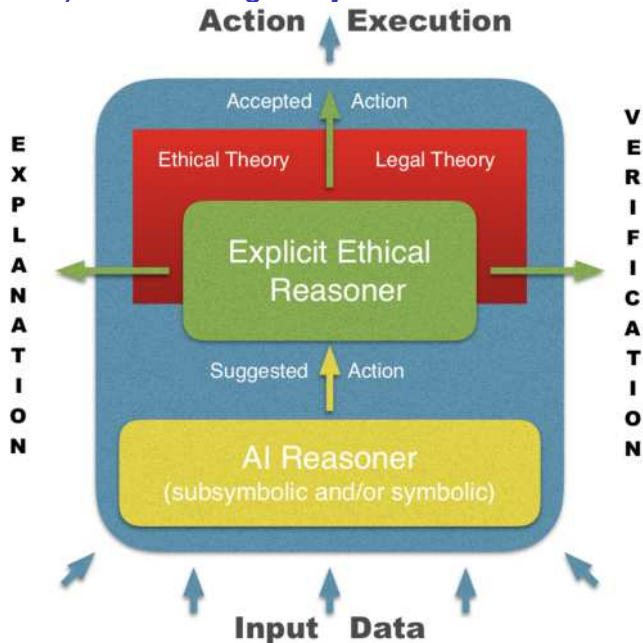
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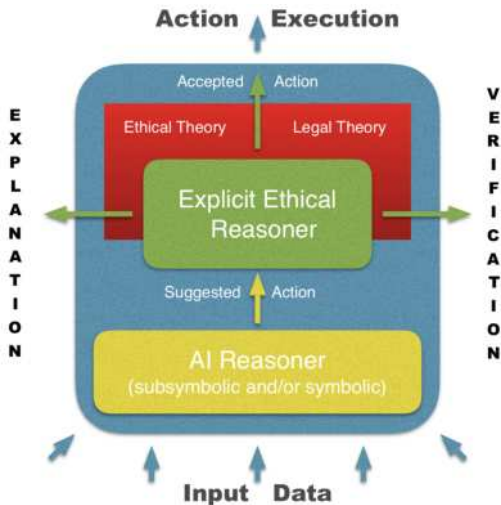
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(Pseudo-)Ethical Intelligent Systems



Related Work

- ▶ Artificial Moral Agents
 - ▶ [Wallach&Allen, 2008]
- ▶ Ethical Governors
 - ▶ [ArkinEtAl., 2009, 2012]
 - ▶ [Dennis&Fisher, 2017]
- ▶ Ethical Deliberation in ART
 - ▶ [Dignum, 2017]
- ▶ Programming Machine Ethics
 - ▶ [Pereira&Saptawijaya, 2016]
- ▶ ...

(Pseudo-)Ethical Intelligent Systems

Address requests towards **Thrustworthy & Responsible AI**

- ▶ **German National AI Strategy (Nov 2018)**

https://www.bmbf.de/files/Nationale_KI-Strategie.pdf

- ▶ **Recommendations from the EU HLEG group (May & June 2019)**



<https://ec.europa.eu/digital-single-market/en/high-level-expert-group-artificial-intelligence>

- ▶ **Ben Goertzel (CEO SingularityNET; Nov 2018): “Toward Democratic, Lawful Citizenship for AIs, Robots, and Corporations”**

<https://tinyurl.com/y8h94ouv>

Which Reasoning Formalisms?

“If we had it [a *characteristica universalis*], we should be able to reason in metaphysics and morals in much the same way as in geometry and analysis.”

(Leibniz, 1677)

Challenges for Explicit Ethical Reasoning Engines: Which Logic(s)?

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Standard CTD structure (Chisholm)

1. obligatory 'a'
 2. obligatory 'if *a* then not *b*'
 3. if 'not *a*' then obligatory '*b*'
 4. 'not *a*'
- (in a given situation)

Danger: Paradox/inconsistency — ex falso quodlibet!

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CTD example (X. Parent): **EU General Data Protection Regulation (GDPR)**

1. Personal data shall be processed lawfully. (Art. 5)
E.g., the data subject must have given consent to the processing. (Art. 6/1.a)
2. **Implicit:** The data shall be kept, for the agreed purposes, if processed lawfully.
3. If personal data has been processed unlawfully, the controller has the obligation to erase the personal data in question without delay. (Art. 17.d, right to be forgotten)
4. **Given situation:** Some personal data has been processed unlawfully.

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Deontic Logic

- ▶ Reasoning about obligations and permissions
- ▶ Two groups of approaches:
 - Possible worlds
 - ▶ standard deontic logic CTD: **no**
 - ▶ dyadic deontic logic CTD: **yes**
 - Norm-based semantics
 - ▶ input/output logic CTD: **yes**



L. van der Torre



X. Parent

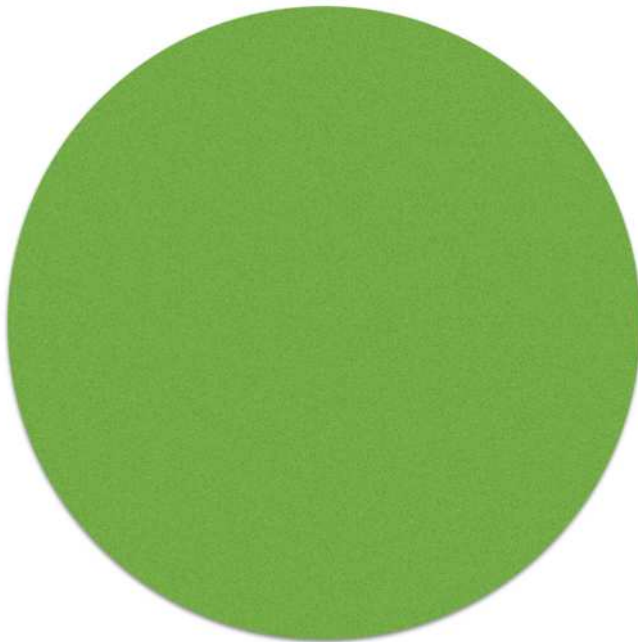


A. Farjami

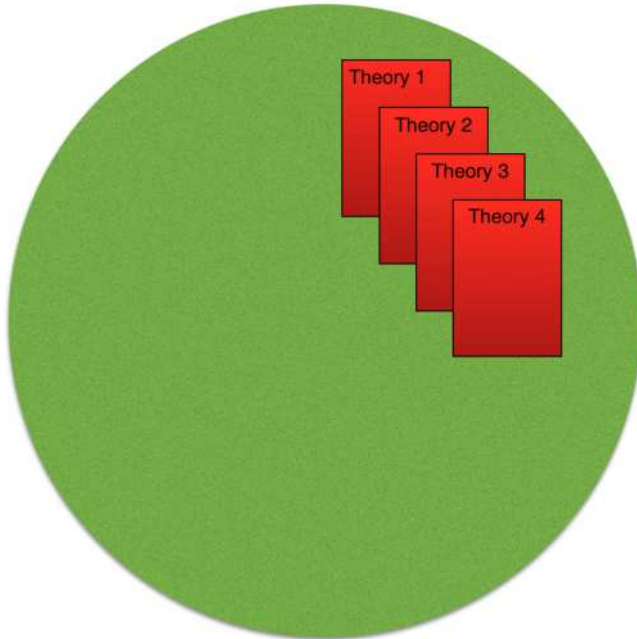
Further interests and challenges

- ▶ Combination with other logics (other modalities)
- ▶ Propositional deontic logic(s) will hardly be sufficient in practice

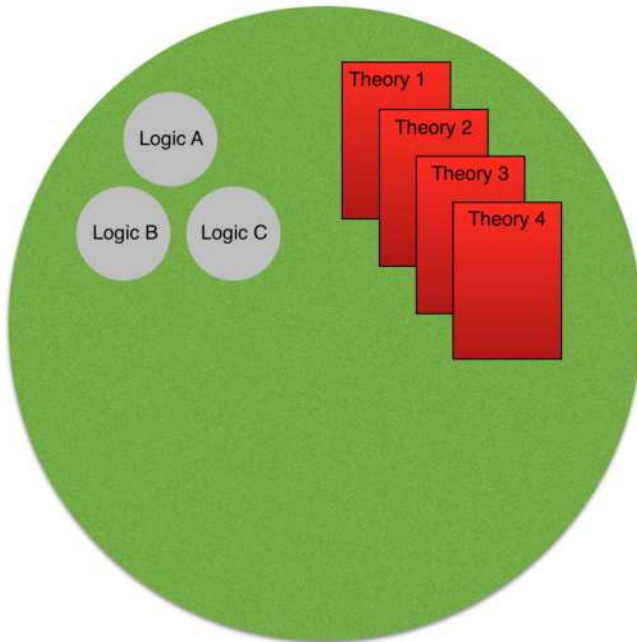
Normative Reasoning Experimentation Platform



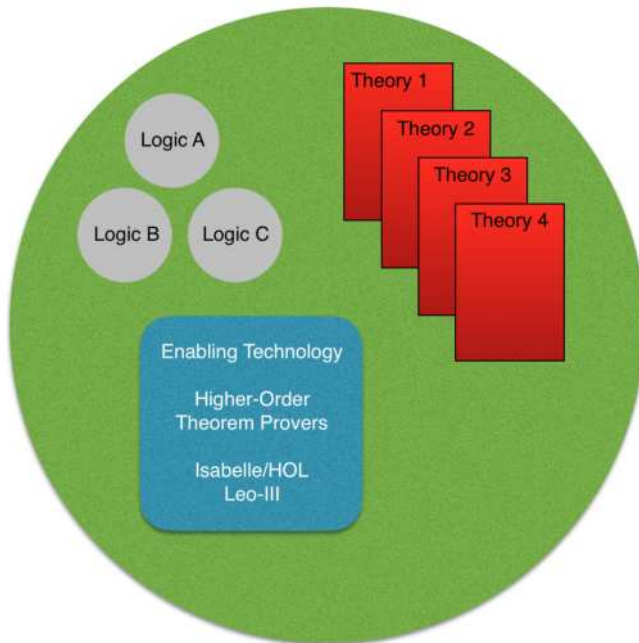
Normative Reasoning Experimentation Platform



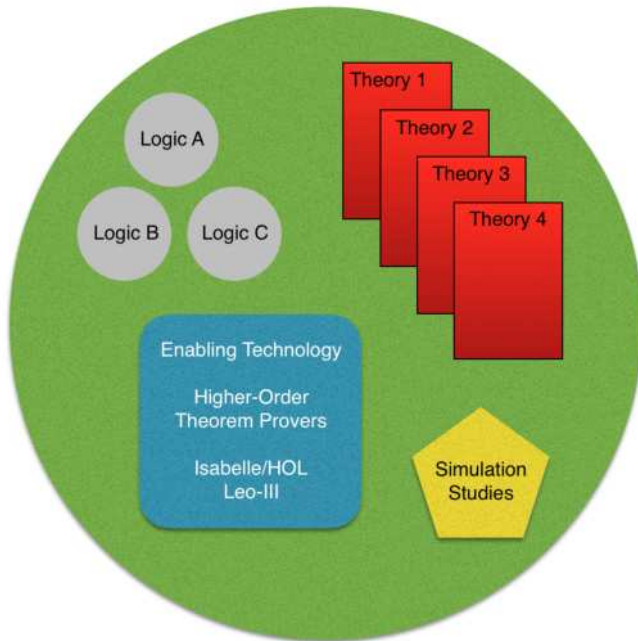
Normative Reasoning Experimentation Platform



Normative Reasoning Experimentation Platform



Normative Reasoning Experimentation Platform

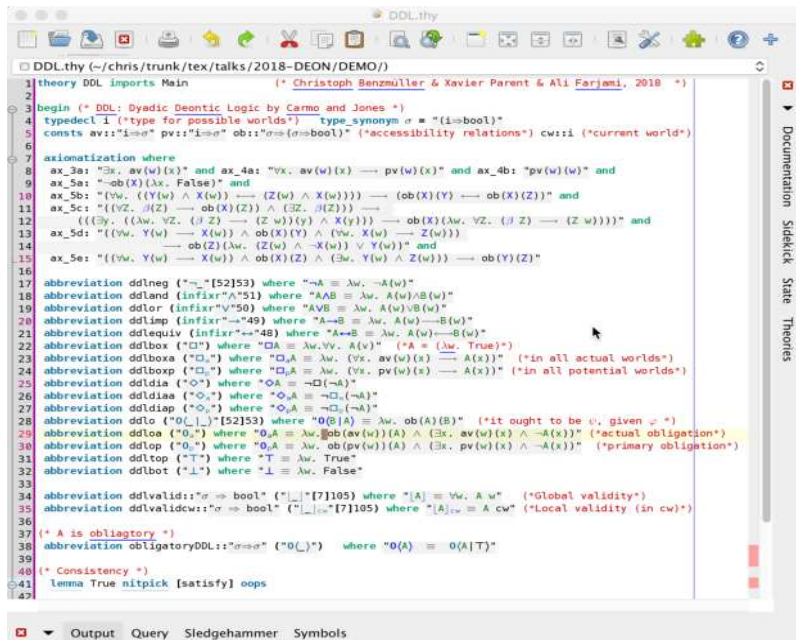


```

1 theory SDL imports Main                                (* Christoph Benzmüller & Xavier Parent, 2018 *)
2
3 begin (* SDL: Standard Deontic Logic (Modal Logic D) *)
4 typedef 1 (*type for possible worlds*) type_synonym  $\sigma$  = "( $i \Rightarrow \text{bool}$ )"
5 consts r::" $i \Rightarrow i \Rightarrow \text{bool}$ " (*Accessibility relation.*) cw::1 (*Current world.*)
6
7 abbreviation mtop ("T") where "T  $\equiv \lambda w. \text{True}$ "
8 abbreviation mbot ("⊥") where "⊥  $\equiv \lambda w. \text{False}$ "
9 abbreviation mnot ("¬") [52,53] where "¬ $\varphi \equiv \lambda w. \neg \varphi(w)$ "
10 abbreviation mand ("∧") [51] where " $\varphi \wedge \psi \equiv \lambda w. \varphi(w) \wedge \psi(w)$ "
11 abbreviation mor ("∨") [50] where " $\varphi \vee \psi \equiv \lambda w. \varphi(w) \vee \psi(w)$ "
12 abbreviation mimp ("→") [49] where " $\varphi \rightarrow \psi \equiv \lambda w. \varphi(w) \rightarrow \psi(w)$ "
13 abbreviation mequ ("↔") [48] where " $\varphi \leftrightarrow \psi \equiv \lambda w. \varphi(w) \leftrightarrow \psi(w)$ "
14 abbreviation obligatory ("OB") where "OB  $\varphi \equiv \lambda w. \forall v. w \ r \ v \ \longrightarrow \ \varphi(v)$ " (*obligatory*)
15 abbreviation mpermissible ("PE") where "PE  $\varphi \equiv \neg(\text{OB } \neg \varphi)$ " (*permissible*)
16 abbreviation mimpermissible ("IM") where "IM  $\varphi \equiv \text{OB } \neg \varphi$ " (*impermissible*)
17 abbreviation omissible ("OM") where "OM  $\varphi \equiv \neg(\text{OB } \varphi)$ " (*omissible*)
18 abbreviation moptional ("OP") where "OP  $\varphi \equiv (\neg(\text{OB } \varphi) \wedge \neg(\text{OB } \neg \varphi))$ " (*optional*)
19
20 abbreviation ddlvalid::" $\sigma \Rightarrow \text{bool}$ " ("⊨") [7,105] (*Global Validity*)
21 where " $\llbracket A \rrbracket \equiv \forall w. A \ w$ "
22 abbreviation ddlvalidcw::" $\sigma \Rightarrow \text{bool}$ " ("⊨cw") [7,105] (*Local Validity (in cw)*)
23 where " $\llbracket A \rrbracket_{cw} \equiv A \ cw$ "
24
25 (* The D axiom is postulated *)
26 axiomatization where D: " $\neg \{(\text{OB } \varphi) \wedge (\text{OB } \neg \varphi)\}$ "
27
28 (* Meta-level study: D corresponds to seriality *)
29 lemma " $\neg \{(\text{OB } \varphi) \wedge (\text{OB } \neg \varphi)\} \longleftrightarrow \{ \forall w. \exists v. w \ r \ v \}$ " by auto
30
31 (* Standardised syntax: unary operator for obligation in SDL *)
32 abbreviation obligatorySDL::" $\sigma \Rightarrow \sigma$ " ("O(_)") where "O(A)  $\equiv \text{OB } A$ "
33
34 (* Consistency *)
35 lemma True nitpick [satisfy] oops

```

Demo: DDL in Isabelle/HOL



```
1 theory DDL imports Main (* Christoph Benzmüller & Xavier Parent & Ali Farjami, 2018 *)
2
3 begin (* DDL: Dyadic Deontic Logic by Carmo and Jones *)
4 typedef 1 (*type for possible worlds*) type_synonym  $\sigma$  = "( $i \Rightarrow \text{bool}$ )"
5 consts av::" $i \Rightarrow \sigma$ " pv::" $i \Rightarrow \sigma$ " ob::" $\sigma \Rightarrow (\sigma \Rightarrow \text{bool})$ " (*accessibility relations*) cw::i (*current world*)
6
7 axiomatization where
8   ax_3a: " $\exists x. \text{av}(w)(x)$ " and ax_4a: " $\forall x. \text{av}(w)(x) \longrightarrow \text{pv}(w)(x)$ " and ax_4b: " $\text{pv}(w)(w)$ " and
9   ax_5a: " $\neg \text{ob}(X)(\lambda x. \text{False})$ " and
10  ax_5b: " $(\forall w. ((\lambda y. \text{av}(w)(y) \longrightarrow (\lambda z. \text{ob}(X)(z) \longrightarrow \text{ob}(X)(y)))) \longrightarrow (\text{ob}(X)(Y) \longrightarrow \text{ob}(X)(Z)))$ " and
11  ax_5c: " $((\forall Z. \beta(Z) \longrightarrow \text{ob}(X)(Z)) \wedge (\exists Z. \beta(Z))) \longrightarrow$ 
12     $((\exists y. ((\lambda w. \forall Z. (\beta(Z) \longrightarrow (\lambda z. \text{ob}(X)(z) \longrightarrow \text{ob}(X)(y) \wedge \text{X}(y))) \longrightarrow \text{ob}(X)(\lambda w. \forall Z. (\beta(Z) \longrightarrow (\lambda z. \text{ob}(X)(z) \longrightarrow \text{ob}(X)(y) \wedge \text{X}(y))))$ " and
13  ax_5d: " $((\forall w. Y(w) \longrightarrow X(w)) \wedge \text{ob}(X)(Y) \wedge (\forall w. X(w) \longrightarrow Z(w))) \longrightarrow \text{ob}(Z)(\lambda w. (\lambda z. \text{ob}(X)(z) \wedge \text{X}(w)) \vee Y(w))$ " and
14  ax_5e: " $((\forall w. Y(w) \longrightarrow X(w)) \wedge \text{ob}(X)(Z) \wedge (\exists w. Y(w) \wedge Z(w))) \longrightarrow \text{ob}(Y)(Z)$ "
15
16 abbreviation ddneg (" $\neg$ "[52]53) where " $\neg A \equiv \lambda w. \neg A(w)$ "
17 abbreviation ddland (infixr" $\wedge$ "51) where " $A \wedge B \equiv \lambda w. A(w) \wedge B(w)$ "
18 abbreviation ddlor (infixr" $\vee$ "50) where " $A \vee B \equiv \lambda w. A(w) \vee B(w)$ "
19 abbreviation ddlimp (infixr" $\longrightarrow$ "49) where " $A \longrightarrow B \equiv \lambda w. A(w) \longrightarrow B(w)$ "
20 abbreviation ddlequiv (infixr" $\longleftrightarrow$ "48) where " $A \longleftrightarrow B \equiv \lambda w. A(w) \longleftrightarrow B(w)$ "
21 abbreviation ddibox (" $\Box$ "[52]53) where " $\Box A \equiv \lambda w. \forall v. A(v)$ " (* $A = (\lambda w. \text{True})$ *)
22 abbreviation ddiboxa (" $\Box_a$ "[52]53) where " $\Box_a A \equiv \lambda w. (\forall x. \text{av}(w)(x) \longrightarrow A(x))$ " (*in all actual worlds*)
23 abbreviation ddiboxp (" $\Box_p$ "[52]53) where " $\Box_p A \equiv \lambda w. (\forall x. \text{pv}(w)(x) \longrightarrow A(x))$ " (*in all potential worlds*)
24 abbreviation ddldia (" $\Diamond$ "[52]53) where " $\Diamond A \equiv \neg \Box(\neg A)$ "
25 abbreviation ddldiaa (" $\Diamond_a$ "[52]53) where " $\Diamond_a A \equiv \neg \Box_a(\neg A)$ "
26 abbreviation ddldiap (" $\Diamond_p$ "[52]53) where " $\Diamond_p A \equiv \neg \Box_p(\neg A)$ "
27 abbreviation ddlo (" $\Box_{\perp}$ "[52]53) where " $\Box_{\perp} A \equiv \lambda w. \text{ob}(A)(B)$ " (*it ought to be  $\psi$ , given  $\psi$ *)
28 abbreviation ddtoa (" $\Box_{\perp}$ "[52]53) where " $\Box_{\perp} A \equiv \lambda w. \text{ob}(\text{av}(w))(A) \wedge (\exists x. \text{av}(w)(x) \wedge \neg A(x))$ " (*actual obligation*)
29 abbreviation ddtop (" $\Box_{\perp}$ "[52]53) where " $\Box_{\perp} A \equiv \lambda w. \text{ob}(\text{pv}(w))(A) \wedge (\exists x. \text{pv}(w)(x) \wedge \neg A(x))$ " (*primary obligation*)
30 abbreviation ddtop (" $\Box_{\perp}$ "[52]53) where " $\Box_{\perp} A \equiv \lambda w. \text{True}$ "
31 abbreviation ddlbot (" $\perp$ "[52]53) where " $\perp \equiv \lambda w. \text{False}$ "
32
33
34 abbreviation ddvalid::" $\sigma \Rightarrow \text{bool}$ " (" $\Box$ "[7]105) where " $\Box A \equiv \forall w. A w$ " (*Global validity*)
35 abbreviation ddvalidcw::" $\sigma \Rightarrow \text{bool}$ " (" $\Box_{\perp}$ "[7]105) where " $\Box_{\perp} A \equiv A cw$ " (*Local validity (in cw)*)
36
37 (* A is obligatory *)
38 abbreviation obligatoryDDL::" $\sigma \Rightarrow \sigma$ " (" $\Box_{\perp}$ "[7]105) where " $\Box_{\perp} A \equiv \Box_{\perp} A$ "
39
40 (* Consistency *)
41 lemma True nitpick [satisfy]oops
42
```

Demo: Normative Reasoning Experimentation Platform

```
1 theory GDPR imports SDL (* Christoph Benzmüller & Xavier Parent, 2018 *)
2
3 begin (** GDPR Example **)
4   consts process_data_lawfully:: $\sigma$  erase_data:: $\sigma$  kill_boss:: $\sigma$ 
5
6   axiomatization where
7     (* It is an obligation to process data lawfully. *)
8     A1: "[0(process_data_lawfully)]" and
9     (* Implicit: It is an obligation to keep the data if it was processed lawfully. *)
10    Implicit: "[0(process_data_lawfully  $\rightarrow$   $\neg$ erase_data)]" and
11    (* If data was not processed lawfully, then it is an obligation to erase the data. *)
12    A2: "[ $\neg$ process_data_lawfully  $\rightarrow$  0(erase_data)]"
13    (* Given a situation where data is processed unlawfully. *) and
14    A3: "[ $\neg$ process_data_lawfully].."
15
16  (** Some Experiments **)
17  lemma True nitpick [satisfy] oops (* Consistency-check: Is there a model? *)
18  lemma False sledgehammer oops (* Inconsistency-check: Can Falsum be derived? *)
19
20  lemma "[0(erase_data)]" sledgehammer nitpick oops (* Should the data be erased? *)
21  lemma "[0( $\neg$ erase_data)]" sledgehammer nitpick oops (* Should the data be kept? *)
22  lemma "[0(kill_boss)]" sledgehammer nitpick oops (* Should the boss be killed? *)
23 end
```

✓ Proof state ✓ Auto update Update Search: 100%

Sledgehammering...
Proof found...
"spass": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could
"e": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could be d
"cvc4": Try this: by (metis A1 A2 A3 D Implicit) (68 ms)
"z3": Try this: by (metis A1 A2 A3 D Implicit) (59 ms)

Output Query Sledgehammer Symbols

Demo: Normative Reasoning Experimentation Platform

The screenshot shows a theorem prover interface with a file named `GDPR.thy` open. The file content is as follows:

```
1 theory GDPR imports SDL (* Christoph Benzmüller & Xavier Parent, 2018 *)
2
3 begin (** GDPR Example **)
4 consts process_data_lawfully::o erase_data::o kill_boss::o
5
6 axiomatization where
7   (* It is an obligation to process data lawfully. *)
8   A1: "[O(process_data_lawfully)]" and
9   (* Implicit: It is an obligation to keep the data if it was processed lawfully. *)
10  Implicit: "[O(process_data_lawfully → ¬erase_data)]" and
11  (* If data was not processed lawfully, then it is an obligation to erase the data. *)
12  A2: "[¬process_data_lawfully → O(erase_data)]"
13  (* Given a situation where data is processed unlawfully. *) and
14  A3: "[¬process_data_lawfully].."
15
16 (*
17  l
18  l
19  l
20  l
21  lemma "[O(¬erase_data)]" sledgehammer nitpick oops (* Should the data be kept? *)
22  lemma "[O(kill_boss)]" sledgehammer nitpick oops (* Should the boss be killed? *)
23 end
```

A red box with a black border is overlaid on the code, containing the text: "Danger Zone: Paradoxes and Inconsistencies!".

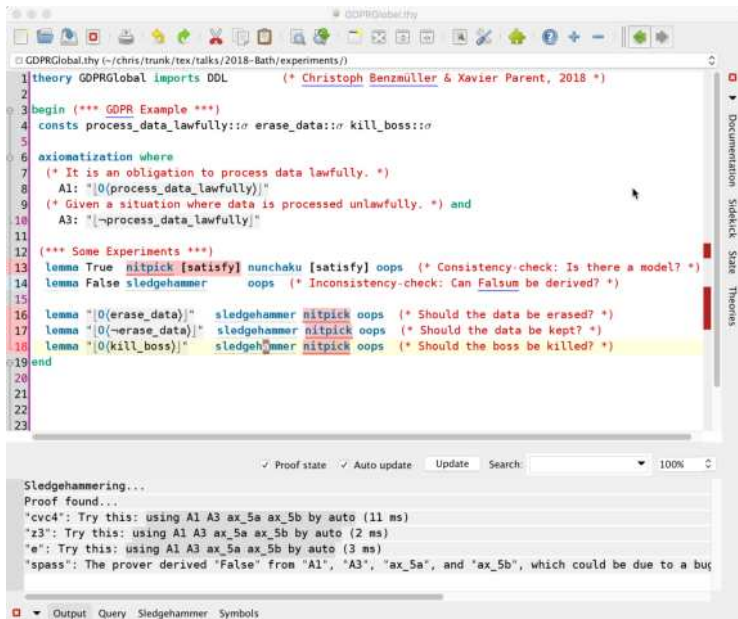
Below the code editor, the interface shows the following options: ☒ Proof state ☒ Auto update Search: 100%

The output area displays the following text:

```
Sledgehammering...
Proof found...
"spass": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could
"e": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could be d
"cvc4": Try this: by (metis A1 A2 A3 D Implicit) (68 ms)
"x3": Try this: by (metis A1 A2 A3 D Implicit) (59 ms)
```

At the bottom, there are tabs for Output, Query, Sledgehammer, and Symbols.

Demo: Global vs. Local Consequence Relation



The screenshot displays the Coq IDE interface. The main editor window shows a Coq script named `GDPRGlobal.thy` with the following content:

```
1 theory GDPRGlobal imports DDL (* Christoph Benzmüller & Xavier Parent, 2018 *)
2
3 begin (** GDPR Example **)
4   consts process_data_lawfully:: $\alpha$  erase_data:: $\alpha$  kill_boss:: $\alpha$ 
5
6   axiomatization where
7     (* It is an obligation to process data lawfully. *)
8     A1: " $\Box$ (process_data_lawfully)"
9     (* Given a situation where data is processed unlawfully. *) and
10     A3: " $\Box$ ( $\neg$ process_data_lawfully)"
11
12   (** Some Experiments **)
13   lemma True nitpick [satisfy] nunchaku [satisfy] oops (* Consistency-check: Is there a model? *)
14   lemma False sledgehammer oops (* Inconsistency-check: Can Falsum be derived? *)
15
16   lemma " $\Box$ (erase_data)" sledgehammer nitpick oops (* Should the data be erased? *)
17   lemma " $\Box$ ( $\neg$ erase_data)" sledgehammer nitpick oops (* Should the data be kept? *)
18   lemma " $\Box$ (kill_boss)" sledgehammer nitpick oops (* Should the boss be killed? *)
19 end
20
21
22
23
```

Below the editor, the `Output` panel shows the execution of the `Sledgehammer` command:

```
Sledgehammering...
Proof found...
"cv4": Try this: using A1 A3 ax_5a ax_5b by auto (11 ms)
"z3": Try this: using A1 A3 ax_5a ax_5b by auto (2 ms)
"e": Try this: using A1 A3 ax_5a ax_5b by auto (3 ms)
"spass": The prover derived "False" from "A1", "A3", "ax_5a", and "ax_5b", which could be due to a bug
```

Demo: Normative Reasoning Experimentation Platform

The screenshot displays a theorem prover interface with a file named `GDPR.thy` open. The file path is `~/chris/trunk/tex/talks/2018-DEON/DEMO/`. The code defines a theory `GDPR` that imports `SDL` and includes a comment: `(* Christoph Benzmüller & Xavier Parent, 2018 *)`. It begins with a section `begin (** GDPR Example **)` and defines constants: `process_data_lawfully::o`, `erase_data::o`, and `kill_boss::o`. An `axiomatization where` block contains three axioms: `A1` (obligation to process data lawfully), `Implicit` (obligation to keep data if processed lawfully), and `A2` (obligation to erase data if not processed lawfully). It also includes `A3` (obligation to kill the boss if data is processed unlawfully). A section `(** Some Experiments **)` follows, containing several lemmas. Lemmas 17 and 18 are consistency and inconsistency checks. Lemmas 20, 21, and 22 are sledgehammer queries. The interface includes a sidebar with `Documentation`, `Sidekick`, `State`, and `Theories`. At the bottom, there are buttons for `Proof state`, `Auto update`, `Update`, and a search bar. The output pane shows the results of a sledgehammer query, indicating that the prover derived "False" from the axioms and providing a proof state.

```
1 theory GDPR imports SDL (* Christoph Benzmüller & Xavier Parent, 2018 *)
2
3 begin (** GDPR Example **)
4   consts process_data_lawfully::o erase_data::o kill_boss::o
5
6   axiomatization where
7     (* It is an obligation to process data lawfully. *)
8     A1: "[0(process_data_lawfully)]" and
9     (* Implicit: It is an obligation to keep the data if it was processed lawfully. *)
10    Implicit: "[0(process_data_lawfully → ¬erase_data)]" and
11    (* If data was not processed lawfully, then it is an obligation to erase the data. *)
12    A2: "[¬process_data_lawfully → 0(erase_data)]"
13    (* Given a situation where data is processed unlawfully. *) and
14    A3: "[¬process_data_lawfully]..".
15
16  (** Some Experiments **)
17  lemma True nitpick [satisfy] oops (* Consistency-check: Is there a model? *)
18  lemma False sledgehammer oops (* Inconsistency-check: Can Falsum be derived? *)
19
20  lemma "[0(erase_data)]" sledgehammer nitpick oops (* Should the data be erased? *)
21  lemma "[0(¬erase_data)]" sledgehammer nitpick oops (* Should the data be kept? *)
22  lemma "[0(kill_boss)]" sledgehammer nitpick oops (* Should the boss be killed? *)
23 end
```

✓ Proof state ✓ Auto update Update Search: 100%

Sledgehammering...
Proof found...
"spass": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could
"e": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could be d
"cvc4": Try this: by (metis A1 A2 A3 D Implicit) (68 ms)
"z3": Try this: by (metis A1 A2 A3 D Implicit) (59 ms)

Output Query Sledgehammer Symbols

Demo: Normative Reasoning Experimentation Platform

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5
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8   A1: "[O(process_data_lawfully)]" and
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12  A2: "[¬process_data_lawfully → O(erase_data)]"
13  (* Given a situation where data is processed unlawfully. *) and
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15
16
17
18
19
20
21 lemma "[O(¬erase_data)]" sledgehammer nitpick oops (* Should the data be kept? *)
22 lemma "[O(kill_boss)]" sledgehammer nitpick oops (* Should the boss be killed? *)
23 end
```

A red box with a black border is overlaid on the code, containing the text:

**Danger Zone:
Paradoxes and Inconsistencies!**

Below the code editor, the interface shows the following information:

- Buttons: ☒ Proof state, ☒ Auto update, Update, Search: [input], 100%
- Text: Sledgehammering...
- Text: Proof found...
- Text: "spass": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could be
- Text: "e": The prover derived "False" from "A1", "A2", "A3", "D", and "Implicit", which could be d
- Text: "cvc4": Try this: by (metis A1 A2 A3 D Implicit) (68 ms)
- Text: "z3": Try this: by (metis A1 A2 A3 D Implicit) (59 ms)
- Buttons: ☒ Output, Query, Sledgehammer, Symbols

“Act in accord with the generic rights of your recipients as well as of yourself. I shall call this the Principle of Generic Consistency (PGC), since it combines the formal consideration of consistency with the material consideration of rights to the generic features or goods of action.”
(Alan Gewirth, Reason and Morality, 1978)

▶ **Gewirth's PGC has**

- ▶ stirred much controversy in moral philosophy
- ▶ been discussed as means to bound the impact of artificial general intellig. (AGI)

▶ **Idea (in a nutshell):**

- ▶ emendation of the Golden Rule (treat others as you would wish to be treated)
- ▶ adopting an agent perspective, the PGC expresses, and deductively justifies, a related upper moral principle, according to which any intelligent agent, by virtue of its self-understanding as an agent, is rationally committed to asserting that:
 - (i) it has rights to freedom and well-being, and
 - (ii) all other agents have those same rights.

▶ **References**

- ▶ A. Gewirth. Reason and morality. U of Chicago Press, 1978.
- ▶ D. Beyleveld. The dialectical necessity of morality: An analysis and defense of Alan Gewirth's argument to the principle of generic consistency. UCP 1991.
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▶ **References**

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- ▶ **References**
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 - ▶ A. Kornai. Bounding the impact of AGI. J. Experimental & Theoretical AI, 2014.

The idea is to constrain potential AGI's to reason in the following way

- ▶ It is necessary for me (as an AGI) to accept that:
 - (P1) I act voluntarily on purpose E (equivalent by definition to "I am a PPA")
 - (C2) E is good (for me)
 - (P3) In order to achieve any purpose whatsoever by my agency, I need my freedom and well-being
 - (C4) My freedom and well-being are necessary goods (for me)
 - (C5) I (even if no one else) have a claim right to my freedom and well-being
- ▶ It is necessary for all PPAs to accept that:
 - (C9) Every PPA has a necessary right to their freedom and well-being

```

15 (** C9: "Every PPA has a necessary right to their freedom and well-being")
16 theorem C9: "[ $\forall a$ . PPA  $a \rightarrow \Box$ . RightTo  $a$  FWB]"
17 proof -
18 {
19   fix I {
20     fix E {
21       (** Stage I *)
22       assume P1: "[ActsOnPurpose I E]" ("I act voluntarily on purpose E")
23       from P1 have P1_var: "[PPA I]" by auto ("definition of PPA")
24       from P1 have C2: "[Good I E]" using explicationGoodness1 by blast ("E is good for me (I)")
25       hence C4: "[ $\Box$ . Good I (FWB I)]" using explicationGoodness2 P3 by blast ("My FWB are necessary goods")
26       (** Stage II *)
27       hence C4a: "[ $\Box$ . FWB I |  $\Box$ . Good I (FWB I)]" using explicationGoodness3 explicationFWB1 by blast
28       hence C4b: "[ $\Box$ . (FWB I)]" using explicationFWB1 explicationFWB2 C4 CJ_14p by blast
29       hence C4c: "[ $\Box$ . ( $\Box$ . (FWB I))]" using OIOAC by auto
30       hence C5a: "[ $\Box$ . ( $\forall a$ .  $\neg$ InterferesWith  $a$  (FWB I))]" using explicationInterference2 by auto
31       hence C5: "[RightTo I FWB]" by simp ("I have a claim right to my freedom and well-being")
32       hence C5_var: "[ $\Box$ . RightTo I FWB]" by simp
33     }
34     (** Stage IIIa *)
35     hence C6: "[ActsOnPurpose I E  $\rightarrow \Box$ . RightTo I FWB]" by (rule impI)
36   }
37   hence C7: "[ $\forall P$ . ActsOnPurpose I P  $\rightarrow \Box$ . RightTo I FWB]" by (rule allI)
38 }
39 hence C8: "[ $\forall a$ .  $\forall P$ . ActsOnPurpose  $a$  P  $\rightarrow \Box$ . RightTo  $a$  FWB]" by (rule allI)
40 hence C9_var: "[ $\forall a$ . PPA  $a \rightarrow \Box$ . RightTo  $a$  FWB]"
41   by simp ("Every PPA has a necessary right to their freedom and well-being")
42 thus ?thesis by simp
43 qed
44

```

☒ Proof state
 ☒ Auto update
 Update
 Search:
 100%

```

proof (prove)
goal {1 subgoal}:
1. ( $\lambda x$ . [PPA  $x$ ])  $\sqsubseteq$  ( $\lambda x$ . pv aw  $\sqsubseteq$  0. ( $\lambda w$ .  $\forall xa$ . ( $\neg$ InterferesWith  $xa$  (FWB  $x$ )) w))

```

☒ Output
 Query
 Sledgehammer
 Symbols

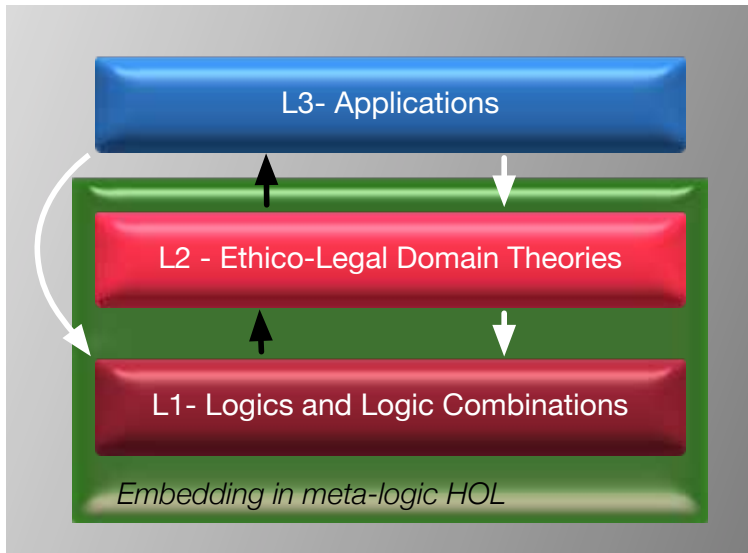
By David Fuenmayor, cf. <http://christoph-benzmueller.de/papers/2018-GewirthArgument.zip>

- ▶ challenging: **alethic & deontic modalities, quantification and indexicals**
- ▶ uninterpreted predicate/relation symbols: FWB, Good (1-ary); ActsOnPurpose, InterferesWith (both 2-ary); NeedsForPurpose (3-ary)
- ▶ $PPA\ a := \exists E. ActsOnPurpose\ a\ E$; additional axiom postulates that being a PPA is identity-constitutive for any individual: $\lfloor \forall a. PPA\ a \rightarrow \Box^D(PPA\ a) \rfloor^D$.
- ▶ $\lfloor \varphi \rfloor^D$ is indexical validity of φ : true in all contexts (following Kaplan's LD)
- ▶ $RightTo\ a\ \varphi := \mathbf{O}_i(\forall b. \neg InterferesWith\ b\ (\varphi\ a))$; a has (claim) right to property φ iff it is obligatory that every b does not interfere with state of affairs $(\varphi\ a)$
- ▶ $\mathbf{O}_i$ is *ideal* obligation operator from Carmo & Jones' DDL ($\mathbf{O}_a$ also fine)
- ▶ further axioms interrelate the concepts of goodness and agency
- ▶ axiom $\lfloor \forall P. \forall a. NeedsForPurpose\ a\ FWB\ P \rfloor^D$ expresses that FWB is always required in order to act on any purpose
- ▶ FWB is postulated a contingent property: $\lfloor \forall a. \Diamond_p FWB\ a \wedge \Diamond_p \neg FWB\ a \rfloor^D$
- ▶ Note that both **first-order** and **higher-order quantifiers** are required

Demo: Gewirth's PGC in Isabelle/HOL [Fuenmayor&B., PRICAI & FroCoS, 2019]

LogiKEY Design and Engineering Methodology

[BenzmüllerParentTorre, arXiv, 2019]



What's Next? — ENORMOUS Plans



What's Next? — ENoRMous Plans



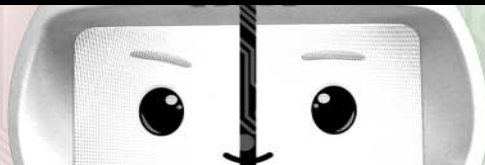
Submitted Proposal: **ENoRME AI-Laboratory**

- ▶ **“Experiments in Normative Reasoning an Machine-Ethics”**
 - Cloud-based Plattform for Universal Normative Reasoning
 - Experiments: Ethico-legal Governance of Autonomous Cars, Financial Trading, Social Robotics, etc.
 - ▶ **Interdisziplinary:** Deduction, Knowledge Representation and Reasoning, Machine Learning, Machine-Ethics, Formal Ethics/Philosophy, Formal Methods
 - ▶ **International Team:** 12 Professors
- Industry Participation**

Ethics

Argued for explicit ethical reasoning competencies in IASs

- ▶ development of normative reasoning experimentation platform
- ▶ utilising HOL as universal meta-logic
- ▶ practical evidence (metaphysics, category theory, etc.)
- ▶ suitable also for teaching



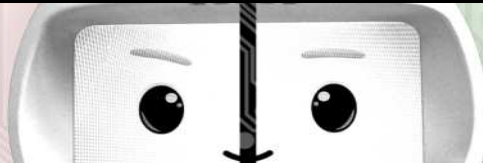
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- ▶ workbench of (expressive) deontic logics and logic combinations
- ▶ formalisation and mechanisation of foundational ethical theories
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